Jaiden Medina

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EDUCATION

Indiana University, Luddy School of Informatics, Computing, & Engineering, Bloomington, IN May 2027
Bachelor of Science in Computer Engineering
Minors: Intelligent Systems Engineering, Mathematics

- Jackson Intelligent Systems Scholarship, Luddy School of Informatics, Computing, & Engineering
- GROUPS Scholar
- Accelerated Master's Program Student in Intelligent Systems Engineering

EXPERIENCE

1st Source Bank, South Bend, IN May 2025-Aug 2025
Information Technology Intern

- Queried the bank's data core using SQL, transformed the data, and developed a Power BI report analyzing rapid account closured; presented findings to top bank leadership to inform retention strategies.
- Assisted in enterprise system migrations, contributing to the successful rewrite and transition of ~10 reports, while gaining hands-on experience with SQL, IBM Cognos, and Hyperion EPM in large scale IT operations.

Lippert, Elkhart, IN
LEAN / Manufacturing Engineer Intern May 2024-Aug 2024

- Led the design, fabrication, installation, and testing of the first-of-its-kind Upper Deck chassis in the RV industry, achieving a ~34% increase in production efficiency and significantly improving safety; pitched the design as a formal product launch to company leadership.
- Programmed a custom calculator using Python, Excel, AutoCAD, and DraftSight, integrated with the production material database, to standardize engineering calculations and streamline workflows across modified rigs.

ACADEMIC PROJECTS

- **Radiation-Induced Fault Characterization in Sequential Logic:** Simulated a 4-bit CMOS master-slave D flip-flop shift register in LTspice; modeled heavy-ion effects via transient current injection to analyze SEU/SET behavior, node sensitivity, and clock-dependent vulnerabilities in digital hardware.

- **Embedded Neural Network Optimization:** Optimized a Raspberry Pi-based neural network for human activity recognition, improving accuracy from ~0.80 to ~0.95 and balancing inference speed, model size, and performance using TensorFlow Lite for edge deployment.

- **Gas Adsorption Physics Research:** Conducted experimental research on helium confinement in MCM-41 mesoporous silica under IU's GROUPS Summer STEM initiative, synthesizing materials, analyzing structural properties, and co-authoring a formal paper contributing to low-dimensional quantum physics.

TECHNICAL

- Programming/Development: Python, C, C++, Java, Verilog, SQL, Assembly, RISC-V, scikit-learn, TensorFlow, Keras, TensorFlow Lite, REST APIs, Swagger/OpenAPI, DAX, YAML, Program Optimization
- Tools/Environments: Microsoft Power BI, IBM Cognos, Hyperion EPM, GitHub, Jetsream2 (Linux VMs), AutoCAD, DraftSight, Autodesk Fusion 360, Microsoft 365 (Excel, Access, SharePoint), Adobe Suite, Arduino

ADDITIONAL

Society of Hispanic Professional Engineers IUB Chapter, Bloomington, IN
Founder & President Aug 2024-Present

- Lead strategy, operations, and executive board for campus chapter focused on underrepresented students in STEM
- Built partnerships with 20+ organizations; led events engaging 500+ students (e.g., Minorities in STEM Night)
- Awarded *Collaboration Powerhouse Award* (Indiana University Student Government) for cross-campus impact